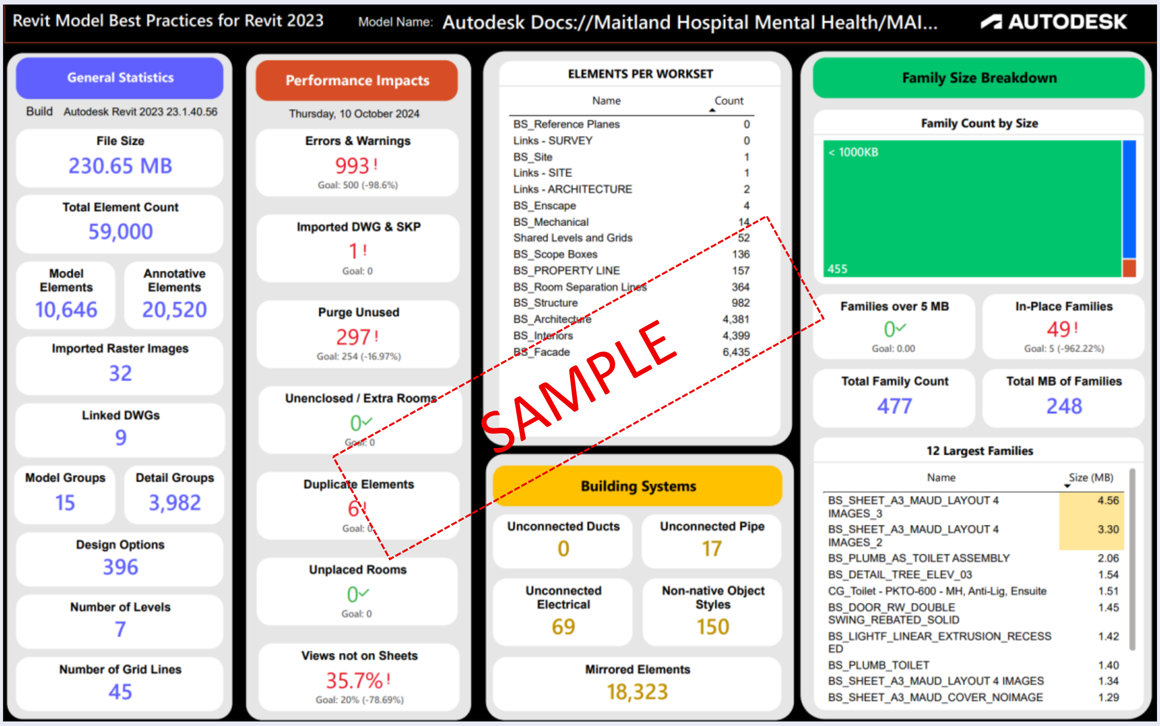
Project BIM Brief

Project Name	[Text]	Project no.	[Text]
ACC Platform Used	**Yes/No	Stage	[Insert phase]
Revit Version	**YEAR	BIM Team Member	**Name
Project Leader	[Text]		

This document is designed to outline efficient and most effective modelling strategies and requirements for projects transitioning or starting phase DD. It serves as a quick reference for reviewing projects based on our BIM standards and protocols which also aides in onboarding new team members by providing a clear understanding of model strategy, processes and expectations.

Model - Previous	Contains (Elements)	
Example Text - *“SURFERSPARADISE_BS_ARCH_ R2023”*	*Example Text* - • *Ground retail and above ground carpark* • *Base build and structural elements* • *2 Resi towers* • *Level 6 Communal, Pools and Amenities* • *Façade elements* • *Cores and stairs* • *All views, sheets, schedules and legends*	

Model Health Check (Optional)



Bates Smart Standard Typical Split


TIER 1 SPLIT STRATEGY - SMALL SCALE WITH 1 BUILDING/TOWER
ARCHITECTURE
&
INTERIORS
SITE MODEL
FACADE MODEL
ARCH & FITOUT MODEL
DWG MODEL
TIER 2 SPLIT STRATEGY - MEDIUM COMPLEXITY WITH 2 BUILDINGS
ARCHITECTURE
COMBINED SCOPE
INTERIORS
SITE MODEL
FACADE MODELS
ARCH MODELS
DWG MODEL
FITOUT MODEL
TIER 3 SPLIT STRATEGY - HIGH COMPLEXITY WITH 3+ BUILDINGS
ARCHITECTURE
COMBINED SCOPE
INTERIORS
CONTROL & SITE MODEL
FACADE MODELS
ARCH MODELS
DWG MODEL
FITOUT MODELS
TIER 1 SPLIT STRATEGY - SMALL SCALE WITH 1 BUILDING/TOWER
ARCHITECTURE
&
INTERIORS
SITE MODEL
FACADE MODEL
A11
ARCH & FITOUT MODEL
A04
A00, A01
A03 - A05
A08 - A10
A12
A15, A18
A20
A22
A30, A40, A50
A06 - A07
A13 - A14
A16
A19
DWG MODEL
TIER 2 SPLIT STRATEGY - MEDIUM COMPLEXITY WITH 2 BUILDINGS
ARCHITECTURE
COMBINED SCOPE
INTERIORS
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A15, A18
A20
A22
A30, A40, A50
DWG MODEL
FITOUT MODEL
A06 - A07
A13 - A14
A16
A19
SAMPLE
SAMPLE

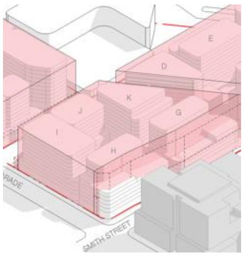
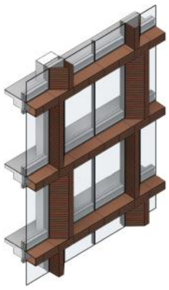
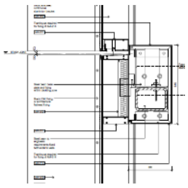
Model Split Strategy	Contains	Comments
Errors & Outcomes	Description	Comments
Shared Coordinates	Description	
North Orientation	Description	
Sheet Strategy	Description	
BEP Considerations & Expectations	Description	Comments

Appendix A: LOD Definitions

Level of Development (LOD)

The Level of Development (LOD) describes the information and geometry of elements within the model that may be utilized through the authorized uses of the model at the current phase of the project. LOD's allow for an understanding of model content requirements and the reliability of the content is established and understood by all stakeholders.

Fundamental LOD Definitions

Level #	Program Stage		Definition	Included In Scope
LOD 100 (Conceptual)	Concept Design		The Model Element may be graphically represented in the Model with a symbol or other generic representation but does not satisfy the requirements for LOD 200. Information related to the Model Element (i.e. cost per square foot, tonnage of HVAC, etc.) can be derived from other Model Elements.	✓
LOD 200 (Approx. Geometry)	Schematic Design		The Model Element is graphically represented within the Model as a **generic** system, object, or assembly with approximate quantities, size, shape, location, and orientation. Non-graphic information may also be attached to the Model Element.	✓
LOD 300 (Precise Geometry)	Design Development / Tender Documentation *along with* Approved for Construction Model		The Model Element is graphically represented within the Model as a **specific** system, object, or assembly in terms of quantity, size, shape, location, and orientation. Non-graphic information may also be attached to the Model Element.	✓
LOD 400 (Fabrication)	Fabrication Model		The Model Element is graphically represented within the Model as a specific system, object or assembly in terms of size, shape, location, quantity, and orientation with detailing, fabrication, assembly, and installation information. Non-graphic information may also be attached to the Model Element.	×
LOD 500 (As-Built)	As built Model		Model elements are modeled as constructed assemblies, actual and accurate in terms of quantity, size, shape, location and orientation. Non-geometric information will be attached to model elements.	×

LOD 200 – SD Phase

Architectural and Interior Systems

In this phase, the project team is to model the following architectural elements to a level that defines the design intent and that accurately represents the design solution.

- Site plan
- Paving, grades, footpaths, kerbs, gutters, site amenities and other elements that are typically represented on site drawings in the vicinity of the proposed building
- Basic interior and exterior walls including doors, windows and openings
- Interior and exterior soffits, overhangs and sun control elements,
- Parapets and screening elements,
- Architectural precast elements
- Floor, ceiling and roof systems
- Insulation in ceiling systems and floors
- Floor, roof and ceiling slopes
- Soffits, openings, and accessories
- Elevators, lifts, stairs and ramp systems
- Furnishings, fixtures, and equipment if not provided by others and integrated into the architectural model for coordination and document generation.
- Model mechanical, electrical and plumbing items that require architectural space (toilets, sinks etc.), require colour/finish selection, louvers and diffuses, lighting fixtures etc. unless provided by the relevant consulting engineers

LOD 300 – DD / Tender / AFC Phase

Model Content Requirements

The project team shall continue to develop the model created in the previous phase. Models are to be further developed in 3D and elements are suitable for the generation of traditional construction documents and shop drawings. As such, analysis and simulation is authorized for detailed elements and systems.

Architectural and Interior Systems

The project team will model the following architectural elements to a level that defines the design intent and that accurately represents the design solution:

- Spaces shall be defined including accurate net square metres and net volume and holding data for the room names and numbers.
- Exterior Walls and Curtain Walls shall be depicted to the exact height, length, width.
- Roofs shall accurately represent the roofscape, drainage system, major penetrations and roof form (build-ups and materials).
- Interior Partitions shall be depicted to the exact height, length, width to properly reflect wall types. Intelligence shall include this and specification/build-up of the system e.g.70mm stud, 2x10mm plasterboard each face, etc.
- Doors, windows and louvers depicted to represent their actual size, type and location. All doors and windows shall be modeled with the necessary intelligence to produce accurate window and door schedules i.e. intelligence contains; - reference number, location, structural opening, wall thickness, door-set overall size, leaf sizes, vision panel/configuration, fire rating, acoustic rating, frame and door materials and finishes, swing type and hand
- Architectural and FF+E fittings (i.e. toilet room accessories, toilet partitions, grab rails, lockers and display cases) etc. shall be accurately depicted. Specialist joinery shall be accurately depicted where necessary and modeled to cover the accurate built volume.
- Ceilings and Soffits shall be depicted at correct heights, dimensions and with correct materials. Ceiling exposed surface shall be set at correct height and the depth of the ceiling shall reflect the construction build up and include zone for the ceiling support structure. Ceiling grids shall be located in the model and agreed early on in the project.\ Any pelmets to be modelled throughout the levels for c
- Any pelmets are to be modelled at the correct height and extent throughout all levels required as per design intent and for coordination purposes.
- Stairs and Ramps shall be accurately depicted with all components (i.e. architectural treads and risers, handrails, guardrails, nosing’s, etc

— Schedules: From the model extract accurate door, window and ceiling, wall and floor finish schedules indicating the type, materials and finishes used in the construction.

Appendix B: Revit Best Practices

General Rules

The following is a collection of best practices from various sources to be implemented on the project. In general, the following rules will improve performance. It is encouraged by all participants in the project to share their experience to all Revit authors across practices.

Linking, Constraints & Design Options

- Minimize constraints and locking to other Revit objects.
- Arrays can be used to copy and associate objects together. After the array is deployed, performance will be improved by ungrouping the array. You can prevent this by unselecting the Group and Associate option before creating the array.
- Limit the use of Rooms in Design Options.
- Use separate models for large design options, rather than the Design Options tool in Revit.
- Remove Design Options when they are not current to the project.
- Consider using separate models for long term design options, which can linked and unlinked as needed.
- Minimize the number of linked files.
- Do not import DWG files.
- Cleanup DWG files before importing; removing hatches and unnecessary geometry and line work.
- If a DWG is unloaded long term in Manage Links, consider removing it from the project.
- Switch off 2D AutoCAD DWG files in perpendicular views, such as Sections and Elevations. Consider creating a Vertical & horizontal workset for these types of linked DWG files.
- Remove all DWG files prior to issuing of Rvt models. Consider a Revit Plugin such as CASE's 'Delete View, Sheets & Links Revit plug-in to perform this task.
- Upgrade links prior to linking, to reduce memory usage and load time.
- Use Origin to Origin
- Remove unneeded raster images and renderings, to improve performance.
- Large raster images that are reduced, such as logos, should be reduced to a smaller printed size and then imported afterwards.

Modelling

- Minimize geometric detail to only be what will be visible at the chose scale. Invest time to the correct level of complexity into the model.
- Until wall, roof, floor, window, door types can be determined, use generic versions of these elements, which uses less geometry.
- Best practice is for models over 200Mb to be broken into separate models. This works best if files can be unloaded whilst working on the project.
- Remove unnecessary Groups. Purge Groups regularly.
- Make backups and archived versions regularly as Purging objects cannot be recovered.
- Review and fix warnings regularly.
- Masking and/or Filled Regions are not to be used to present a model element used in a 2D detail.

Views

- Before closing a file, keep a simple drafting view set as the Starting View to reduce time when opening. This can include notes for the person who will be opening the file next.
- Minimize view depth where possible in elevation, plan and section. Use View templates to standardize these across the model file.
- Use Section Boxes to limit visible geometry when working in a 3D view.
- Reduce linked models file size to improve performance by deleting views and purging.
- Avoid turning on shadows in views. Turn off shadows before printing unless necessary.
- Use wireframe or Shaded modes, which is faster than Hidden or Shaded with Edges.
- Zoom in to speed up drawing and snapping.
- If you have a very dense view and snap lines appear to select options in all different directions, clear the Snap to Remote Objects option in the Snaps Dialog.

- Close any unnecessary windows/Views that you have open; this will improve performance as they are all loaded into RAM. When Saving-to-Central close all hidden views as it will be regenerated during the saving process.
- Close any unnecessary Views before making major changes of reloading links, as the changes will only need to happen in the open view.
- Associate standard views(plan, section, elevation) to Scope boxes for consistency.
- Workset are to be used and elements are to be placed on the correct workset.

Rooms and Spaces

- Promptly resolve warnings about volume boundaries overlapping.
- Avoid coincident room or space separation lines overlapping each other and overlapping walls. To locate room or space separation lines in a model, create a wireframe view template with walls and room or space separation lines visible.
- Set the colour for room or space separation lines to red with a heavy line weight so they are easy to identify.
- Place room or space separation lines on one workset for better control.
- Turning area and volume computation off can improve performance, but will disable much of the volume analytical functionality in Revit MEP. When volume computation is off, Revit-based applications will represent rooms as simple extrusions, without considering ceilings, roofs, floors, or other upper or lower boundaries. Because volume computations may affect the performance of your Revit project, it is disabled by default, except in Revit MEP, where the setting is enabled by default. Turn volume computations on, when you need.

Grids, Levels

- Associate Grids and Levels to their own Scope boxes for consistency.

Worksets

- The Revit platform team recommends that worksets should be employed most often to segregate conceptual areas of a project, such as:
- Separate buildings
- Grid and levels
- Building core
- Building shell
- Furniture and equipment spanning multiple categories
- Spatially identifiable areas of a single building
- Linked RVT and DWG files
- Room or space separation lines (see Volumes - Rooms and Spaces section)
- More workset information:
- Checking out a workset may occasionally be of use if certain model elements, such as the building grid or linked files, need to be protected from accidental change. In that event, BIM managers or team leaders will sometimes check out a workset containing project elements that should not be casually edited or relocated.
- Worksets can help manage element visibility to reduce visual clutter during editing, as well as the Revit platform's use of memory. Closing currently unneeded worksets can release allocated RAM for the Revit platform's use in memory-intensive tasks such as printing and exporting.
- Although closed worksets will not appear in views, any elements within them will still be updated during model regeneration if they are impacted by changes made in open worksets, thus maintaining data coordination across the model.
- Use selective workset opening when accessing a work shared profile file for reduced opening times.
- Close worksets not required when editing models.
- When creating a workset, clear the 'Visible by default' option sparingly. Use this only for un-built or temporary objects as it will cause issues further into the project.
- Do not use 'workset1', create a more descriptive workset.
- Setup a view to isolate check worksets, such as grids and levels to make sure all objects are on their correct workset.

Worksharing

- Any large project requires constant team coordination. Most customers have realized many benefits by providing teams with instant messaging (IM) software to help coordinate model editing and saving between geographically dispersed teams.

— When making significant changes in a project (moving a level or major geometry changes), it is recommended that you perform the operation when no other users are working on the file and all other users have relinquished all elements. Once the changes are saved to central, have all users make new local files.

— Save to Central operations can be accelerated by a preceding Reload Latest command.

— When a project has been edited by other users for a day or more, it may be faster to create a new local from the central model rather than relying on the Reload Latest command to update the individual local model a day or more behind the remainder of the team.

— Try to keep project team workstation specifications equivalent. A dramatically weaker machine specification used by a single team member can reduce overall project performance.

— Revit-based applications save only one local model at a time to the central file. As deadlines approach and the frequency of saving to central increases, use the Worksharing Monitor feature to coordinate Save to Central commands across the team. Alternatively, team members could stagger their regular saves to central by either communicating an intention to save to central to the team or by assigning a standard saving time to each team member; for example, at ten or fifteen minutes after the hour.

— When attempts to save to central collide, Revit will notify you that another user is currently saving to central. Cancelling the Save to Central operation will prevent queuing the save request, allowing the user to continue to edit the local file before another Save to Central command.

— Because the Revit platform attempts to update all open views before saving, both local saves and saving to central will increase in performance if a simple view, such as a drafting view, is the only view open when the save operation begins.

— To reduce disk usage and memory usage, regularly compact central and local files. TIP When you need to upgrade a workshared file from one version of Revit to another, rename all of the central files first. If you have linked files, Revit cannot find the links and thus will not try to temporarily upgrade the links. After you have upgraded, you can then use the Save As command to revert to the original name. Then, when you upgrade the linked files, Revit will once again find the upgraded linked file.